Voltage dividers

National 5 · Electronics & control · Answers

Answers and marking guidance

Working is shown for each answer. Other valid methods should be credited.

1 Calculate the missing resistance.

- $V_2 = 15$ V
- $R_1 = 9 \div 15 \times 26 = 15.6\ \Omega$

2 Calculate V_2 or V_{out} for each divider.

- a) $15 \div 50 \times 10 = 3$ V
- b) $3300 \div 5500 \times 6 = 3.6$ V
- c) $240 \div 340 \times 22 = 15.5$ V

3 Calculate V_{out} for each divider.

- a) $3300 \div 9900 \times 12 = 4$ V
- b) $20 \div 30 \times 30 = 20$ V

4 Calculate the missing resistance in each divider.

- a) $V_2 = 120$ V; $R = 110 \div 120 \times 2700 = 2.5$ kΩ
- b) $V_2 = 5$ V; $R = 25 \div 5 \times 4300 = 21.5$ kΩ
- c) $V_1 = 3.5$ V; $R = 3.6$ kΩ (equal voltages, so equal resistances)

5 Calculate the missing resistance in each divider.

- a) $V_1 = 3$ V; $R = 3 \div 9 \times 500 = 167\ \Omega$
- b) $V_1 = 188$ V; $R = 188 \div 42 \times 45 = 201\ \Omega$